

An Analysis of Black Friday Sales Data in USA - 2025

```
!pip install pyspark
```

Requirement already satisfied: pyspark in /usr/local/lib/python3.12/dist-packages (4.0.2)

Requirement already satisfied: py4j<0.10.9.10,>=0.10.9.7 in /usr/local/lib/python3.12/dist-packages (from pyspark) (0.10.9.9)

```
from pyspark.sql import SparkSession
```

```
spark = SparkSession.builder \
    .appName("A1. Assignment") \
    .getOrCreate()
```

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) retail_black...es_100k.csv

retail_black_friday_sales_100k.csv(text/csv) - 11135997 bytes, last modified: 5/30/2026 - 100% done
Saving retail_black_friday_sales_100k.csv to retail_black_friday_sales_100k.csv

```
df = spark.read.csv(
    "retail_black_friday_sales_100k.csv",
    header=True,
    inferSchema=True
)
```

```
df.printSchema()
df.show(10)
```

```
root
|-- transaction_id: string (nullable = true)
|-- customer_id: string (nullable = true)
|-- age_group: string (nullable = true)
|-- gender: string (nullable = true)
|-- city: string (nullable = true)
|-- customer_segment: string (nullable = true)
|-- product_id: string (nullable = true)
|-- product_category: string (nullable = true)
|-- original_price: double (nullable = true)
|-- discount_pct: integer (nullable = true)
|-- final_price: double (nullable = true)
|-- quantity: integer (nullable = true)
|-- purchase_amount: double (nullable = true)
|-- payment_method: string (nullable = true)
|-- purchase_date: date (nullable = true)
|-- purchase_hour: integer (nullable = true)
|-- is_weekend: integer (nullable = true)
|-- is_black_friday: integer (nullable = true)
```

transaction_id	customer_id	age_group	gender	city	customer_segment	product_id	product_category	original_price	discount_pct	final_price	quantity	purchase_amount	payment_method	purchase_date	purchase_hour	is_weekend	is_black_friday
T0000001	C007297	26-35	Male	San Francisco	Loyal	P2713	Footwear	153.73									
T0000002	C001640	56+	Other	Dallas	Returning	P7219	Beauty	230.05									
T0000003	C018025	26-35	Male	Phoenix	New	P5521	Groceries	174.67									
T0000004	C016050	18-25	Other	Miami	VIP	P3003	Sports	399.61									
T0000005	C014629	46-55	Male	Los Angeles	Loyal	P4633	Footwear	120.87									
T0000006	C009145	46-55	Other	Seattle	Returning	P4013	Accessories	192.91									
T0000007	C006718	46-55	Female	San Francisco	Returning	P1445	Home & Kitchen	792.57									
T0000008	C005698	46-55	Female	Seattle	Returning	P7501	Footwear	209.35									
T0000009	C027652	36-45	Female	Los Angeles	Returning	P4350	Groceries	114.32									
T0000010	C002083	36-45	Other	San Francisco	New	P6243	Sports	265.56									

only showing top 10 rows

```
df.count()
df.groupBy("city").sum("purchase_amount").show()
```

city	sum(purchase_amount)
Phoenix	3526014.4299999997
Dallas	3547967.4299999994
Los Angeles	3430433.0200000047
San Francisco	3445376.2999999996
Chicago	3512360.6299999997
Atlanta	3618348.5199999996

```

|      Seattle| 3456842.390000004|
|      Houston| 3474957.449999983|
|      Miami| 3566829.800000003|
|      New York| 3546500.69999999|
+-----+

```

Task 2 — Data Cleaning

Identify missing values

```
df.filter(df.purchase_amount.isNull()).count()
```

```
0
```

Task 03 - Exploratory Data Analysis (EDA)

Summary statistics for numerical columns

```
# Summary statistics for numerical columns
df.describe().show()
```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|summary|transaction_id|customer_id|age_group|gender|city|customer_segment|product_id|product_category|original_price|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|count|100000|100000|100000|100000|100000|100000|100000|100000|100000|
|mean|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|330.2819112999994|
|stddev|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|444.3159159152222|
|min|T0000001|C000001|18-25|Female|Atlanta|Loyal|P1000|Accessories|5.0|
|max|T0100000|C035000|56+|Other|Seattle|VIP|P9999|Toys|2499.89|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

```
# GroupBy analysis: Total purchase amount by Gender
```

```
df.groupby("gender").sum("purchase_amount").show()
```

```

+-----+-----+
|gender|sum(purchase_amount)|
+-----+-----+
|Female|1.16389316899999979E7|
|Other|1.1719413020000003E7|
|Male|1.1767285959999994E7|
+-----+-----+

```

```
# Sorting: Top 10 highest purchases
```

```
df.orderBy(df.purchase_amount.desc()).show(10)
```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|transaction_id|customer_id|age_group|gender|city|customer_segment|product_id|product_category|original_price|disc
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|T0006388|C001139|36-45|Male|Dallas|Returning|P3120|Electronics|2408.8|
|T0034867|C012695|18-25|Female|Chicago|VIP|P8291|Electronics|2460.23|
|T0006648|C019630|18-25|Female|Chicago|Returning|P6272|Electronics|2329.8|
|T0017554|C024872|56+|Female|Chicago|Returning|P8605|Electronics|2492.08|
|T0051067|C033147|18-25|Male|Atlanta|Loyal|P6390|Electronics|2415.46|
|T0003042|C013072|36-45|Male|San Francisco|New|P8510|Electronics|2155.87|
|T0074499|C026446|18-25|Female|Dallas|Returning|P2354|Electronics|2393.8|
|T0022435|C020350|26-35|Female|Atlanta|New|P8791|Electronics|2384.33|
|T0027968|C028692|56+|Other|Miami|Returning|P2467|Electronics|2177.11|
|T0041909|C031820|18-25|Other|Los Angeles|Returning|P6377|Electronics|2300.15|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

only showing top 10 rows

```
# Filtering: Analysis of customers in a specific Age group (e.g., 26-35)
```

```
df.filter(df.age_group == "26-35").show()
```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|transaction_id|customer_id|age_group|gender|city|customer_segment|product_id|product_category|original_price|disc
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|T0000001|C007297|26-35|Male|San Francisco|Loyal|P2713|Footwear|153.73|
|T0000003|C018025|26-35|Male|Phoenix|New|P5521|Groceries|174.67|
|T0000012|C006141|26-35|Other|San Francisco|Returning|P8474|Groceries|28.7|
|T0000014|C015248|26-35|Male|Los Angeles|Loyal|P2203|Home & Kitchen|1063.85|
|T0000016|C001740|26-35|Female|Seattle|New|P6387|Sports|295.26|
|T0000018|C027494|26-35|Female|Houston|New|P7967|Groceries|168.35|
|T0000022|C000426|26-35|Female|Phoenix|Returning|P9909|Electronics|1854.46|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

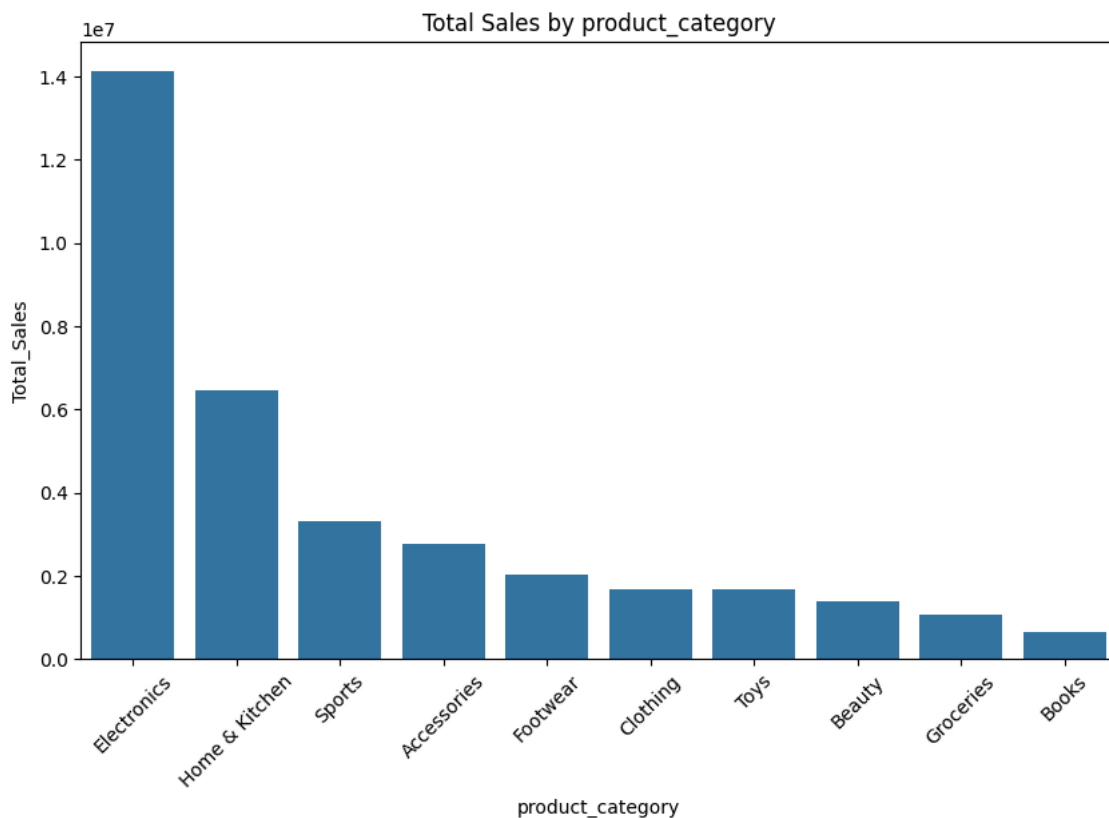
T0000025	C022299	26-35	Female	Houston	New	P3714	Books	45.55
T0000030	C006699	26-35	Male	Houston	Returning	P8777	Beauty	207.35
T0000034	C023527	26-35	Male	Atlanta	Returning	P3465	Home & Kitchen	162.15
T0000038	C030109	26-35	Female	Phoenix	Loyal	P5372	Footwear	106.84
T0000048	C018966	26-35	Other	Phoenix	Returning	P8307	Beauty	90.91
T0000050	C015257	26-35	Male	Miami	Returning	P6482	Groceries	114.24
T0000054	C029715	26-35	Other	New York	New	P3841	Clothing	37.37
T0000059	C013731	26-35	Female	Los Angeles	Returning	P3976	Toys	224.69
T0000064	C010709	26-35	Other	San Francisco	Returning	P5999	Accessories	101.42
T0000067	C017692	26-35	Other	Dallas	Returning	P6640	Beauty	49.0
T0000078	C020623	26-35	Other	Phoenix	Returning	P1796	Books	80.68
T0000079	C013935	26-35	Other	Miami	VIP	P9346	Electronics	646.19
T0000081	C025929	26-35	Other	Seattle	Returning	P2964	Footwear	96.08

only showing top 20 rows

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
product_category_sales_df = spark.sql("SELECT product_category, SUM(purchase_amount) as Total_Sales FROM retail_black_friday_sales")
category_data_pd = product_category_sales_df.toPandas()
```

```
plt.figure(figsize=(10, 6))
sns.barplot(x='product_category', y='Total_Sales', data=category_data_pd)
plt.title('Total Sales by product_category')
plt.xticks(rotation=45)
plt.show()
```



Task 4 — SQL Analysis

```
# Create a temporary view
```

```
df.createOrReplaceTempView("retail_black_friday_sales_100k")
```

```
# 1. Top 5 highest transactions
```

```
spark.sql("SELECT * FROM retail_black_friday_sales_100k ORDER BY purchase_amount DESC LIMIT 5").show()
```

transaction_id	customer_id	age_group	gender	city	customer_segment	product_id	product_category	original_price	discount_price
T0006388	C001139	36-45	Male	Dallas	Returning	P3120	Electronics	2408.8	
T0034867	C012695	18-25	Female	Chicago	VIP	P8291	Electronics	2460.23	1
T0006648	C019630	18-25	Female	Chicago	Returning	P6272	Electronics	2329.8	
T0017554	C024872	56+	Female	Chicago	Returning	P8605	Electronics	2492.08	1

T0051067	C033147	18-25	Male	Atlanta	Loyal	P6390	Electronics	2415.46	1
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2. Average sales by Product ID

```
spark.sql("SELECT product_id, AVG(purchase_amount) as Avg_purchase_amount FROM retail_black_friday_sales_100k GROUP BY product_id")
```

product_id	Avg_purchase_amount
P6976	691.3799999999999
P4634	763.1399999999999
P5384	444.518125
P2949	458.0683333333333
P3024	396.9023809523809
P3122	411.5033333333333
P5100	389.4852941176471
P6207	421.0681250000000
P7769	381.3646666666667
P1670	146.3516666666667
P4489	342.1099999999999
P7303	286.23125
P1311	519.0191666666667
P5306	226.1511111111111
P1815	469.4163636363636
P3826	398.6358333333333
P7806	296.32
P2813	588.6644444444444
P5629	313.8977777777778
P2456	639.9944444444444

only showing top 20 rows

3. City-wise sales summary

```
spark.sql("SELECT city, SUM(purchase_amount) as Total_Sales FROM retail_black_friday_sales_100k GROUP BY city").show()
```

city	Total_Sales
Phoenix	3526014.429999999
Dallas	3547967.429999994
Los Angeles	3430433.020000004
San Francisco	3445376.299999996
Chicago	3512360.629999997
Atlanta	3618348.519999996
Seattle	3456842.390000004
Houston	3474957.449999983
Miami	3566829.800000003
New York	3546500.699999999

Task 5 — Insights and Interpretation

Which product_categories perform best during Black Friday?

```
spark.sql("SELECT product_category, SUM(purchase_amount) as Total_Sales FROM retail_black_friday_sales_100k GROUP BY product_category")
```

product_category	Total_Sales
Electronics	1.4123441469999976E7
Home & Kitchen	6461099.920000016
Sports	3303563.290000004
Accessories	2754561.540000004
Footwear	2013317.390000003
Clothing	1688471.880000001
Toys	1669209.909999997
Beauty	1374751.970000001
Groceries	1075986.120000003
Books	661227.180000002

Which customer_segments spend the most?

```
df.groupBy("customer_segment").sum("purchase_amount").show()
```

customer_segment	sum(purchase_amount)
------------------	----------------------

```
+-----+-----+
|      Loyal| 8888416.699999968|
|      Returning|1.2125507739999983E7|
|      New|1.0574443419999978E7|
|      VIP| 3537262.809999996|
+-----+-----+
```

What time of day drives the most purchases?

```
# Create a temporary view

df.createOrReplaceTempView("black_friday_sales")

# Run a SQL query

spark.sql("""
    SELECT purchase_hour, COUNT(*) as Total_Transactions
    FROM retail_black_friday_sales_100k
    GROUP BY purchase_hour
    ORDER BY Total_Transactions DESC
    LIMIT 1
    """).show()
```

```
+-----+-----+
|purchase_hour|Total_Transactions|
+-----+-----+
|          10|              4295|
+-----+-----+
```